K Sathwik Karanth

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SUMMARY

Final-year Computer Science and Engineering student with a keen interest in Full-Stack Web Development and software development. Enjoy building web applications and learning new technologies related to modern web systems. Strong foundation in programming, databases, and problem-solving concepts. Motivated to contribute to real-world projects while continuously improving technical and professional skills.

PROJECTS

Online Payment Fraud Detection
Jan. 2026 – Present

- Built a fraud detection system using **Random Forest** and **Logistic Regression** on transactional datasets, achieving approximately **97% accuracy**.
- Performed data preprocessing, feature engineering, and handled class imbalance using **SMOTE** to improve fraud detection recall.
- Developed and deployed a **Flask-based web application** for real-time transaction classification as legitimate or fraudulent.

Women Safety Management System using IoT
Sept. 2025 – Nov. 2025

- Developed a wearable safety device using **Arduino, GPS, and GSM modules** to send emergency alerts with live location tracking.
- Implemented SOS and panic-button functionality for instant communication with pre-configured emergency contacts.
- Designed a compact, low-power embedded system focused on reliability and rapid activation during emergencies.

Smart Home Automation System
Jan. 2025 – May 2025

- Designed an IoT-based home automation system using **Arduino, ESP8266, and Blink** for remote control of household appliances.
- Integrated motion, temperature, and light sensors to enable intelligent and automated device switching.
- Implemented secure cloud communication for real-time monitoring and control over Wi-Fi networks.

EDUCATION

A J Institute of Engineering & Technology Aug. 2023 – May 2027
B.E. in Computer Science & Engineering | CGPA: 8.2 Mangalore, Karnataka

SKILLS

Languages: Java, Python, C, HTML, CSS, JavaScript, Dart
Frameworks: Flutter, Flask
Tools: Google Cloud Platform, GitHub, Git, Gradle, Docker, Jenkins, Latex, Arduino
Database: MongoDB, MySQL, SQLite
Concepts: Machine Learning, Artificial Intelligence, Operating System, DSA
Design & Visualization: Figma, Canva, Matplotlib, Seaborn, Plotly

CERTIFICATIONS & EXTRACURRICULAR

- **Tech A Java Developer Certification – Infosys Springboard:** Completed training in Java programming, object-oriented concepts, exception handling, collections, and application development.
- **VLSI Digital Design – Infosys Springboard:** Gained knowledge of digital circuit design, logic gates, combinational and sequential circuits, and semiconductor fundamentals.
- **Generative AI for All – Infosys Springboard:** Learned the fundamentals of Generative AI, Large Language Models (LLMs), prompt engineering, and AI-driven applications.
- **AI & Cloud Cybersecurity: Building Digital Systems Workshop:** Explored the integration of AI, Cloud Computing, and Cybersecurity in modern digital infrastructures; learned security best practices, cloud-based deployment models, and AI-powered threat detection techniques.
- **Internet of Things (IoT) Workshop:** Acquired hands-on experience with IoT architectures, sensors, microcontrollers, and real-time device communication; developed an understanding of smart automation systems and connected device ecosystem